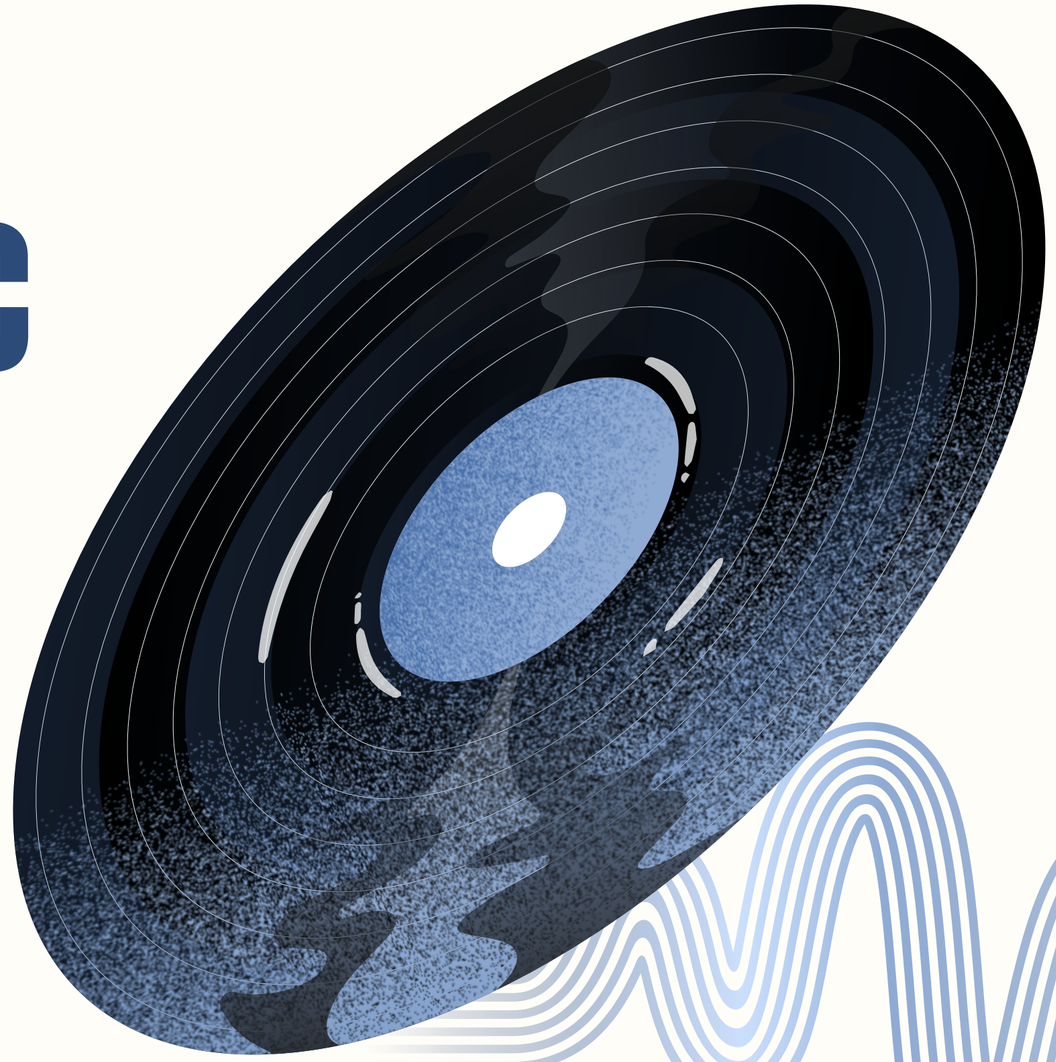
AI VS ORIGINAL MUSIC CLASSIFICATION



Presented by

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Mentor: **Olesia Omelchuk**

ROLES

Alina Bodnar

Handcrafted acoustic feature extraction, exploratory data analysis (EDA), feature group comparison, and feature importance analysis

Olena Dovbenchuk

Dataset preparation, group-aware data splitting, CLAP, MERT and MuQ embeddings, PCA/UMAP visualization, and cross-model error analysis.

Sofiia Vitchynkina

Baseline classification and evaluation using Logistic Regression, SVM, and k-NN across four audio representations and performance evaluation.

DATA & PIPELINE

FakeMusicCaps

27,605
audio clips

5
generators

5,521
unique track IDs



FakeMusicCaps audio
5 generators

↓

Preprocessing
Trim / pad every clip to 10 s

↓

Audio representations

↙ ↓ ↓ ↘

Handcrafted
36 Acoustic features

CLAP embeddings
512 dimension

MERT embeddings
768 dimension

MuQ embeddings
1024 dimension

↓

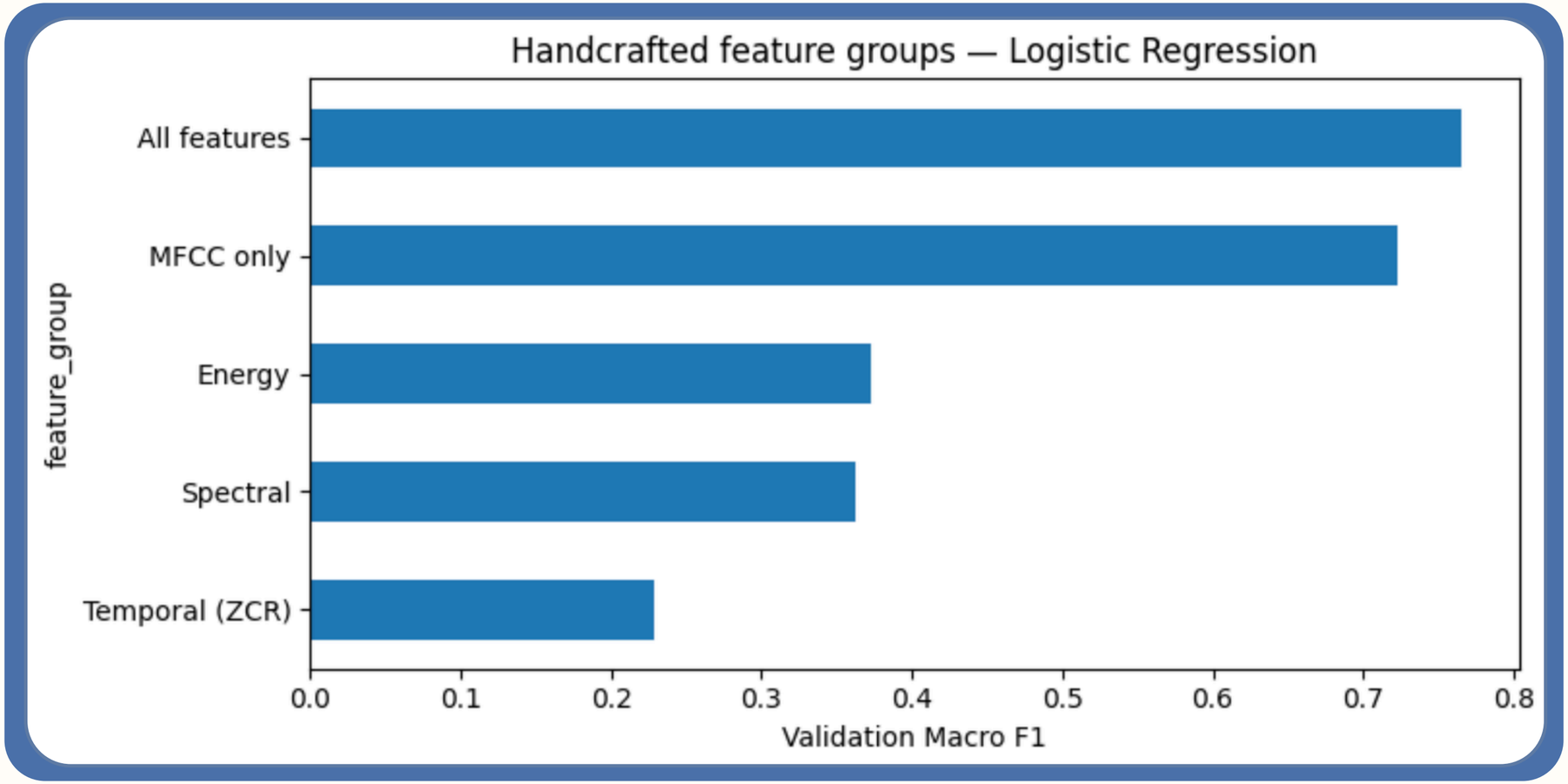
Classification
Logistic Regression · SVM · k-NN

↓

Evaluation
Macro F1 · Balanced Accuracy · Precision · Recall

04

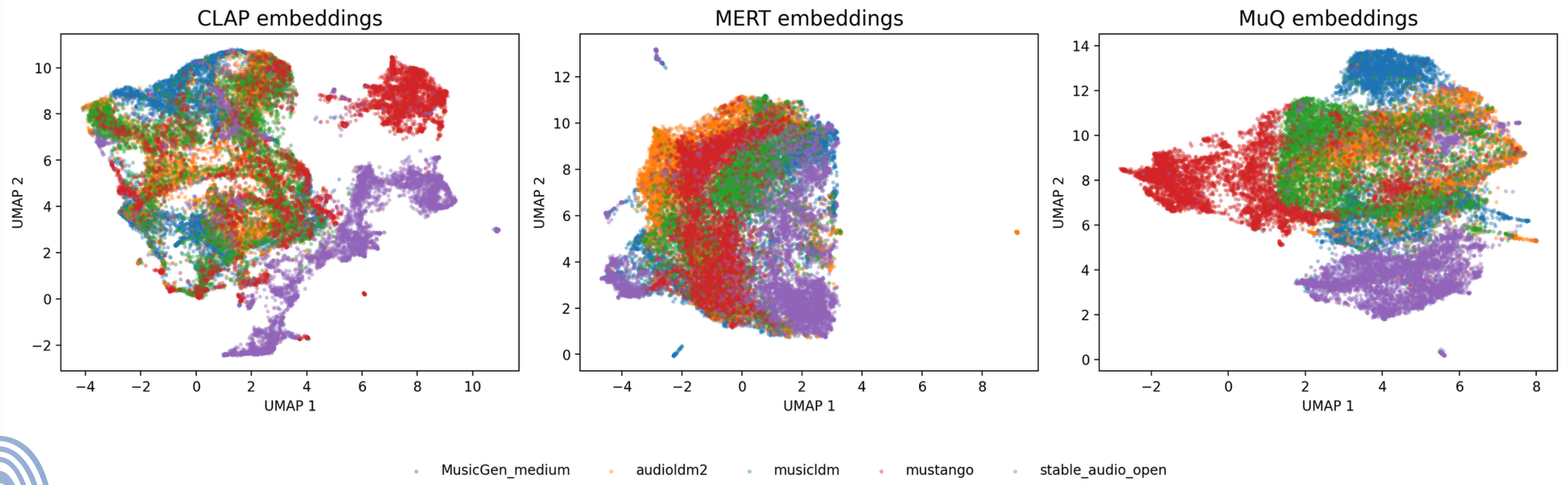
HANDCRAFTED FEATURE



- MFCCs were the strongest individual feature group.
- Combining all handcrafted features gave the best handcrafted result.

05

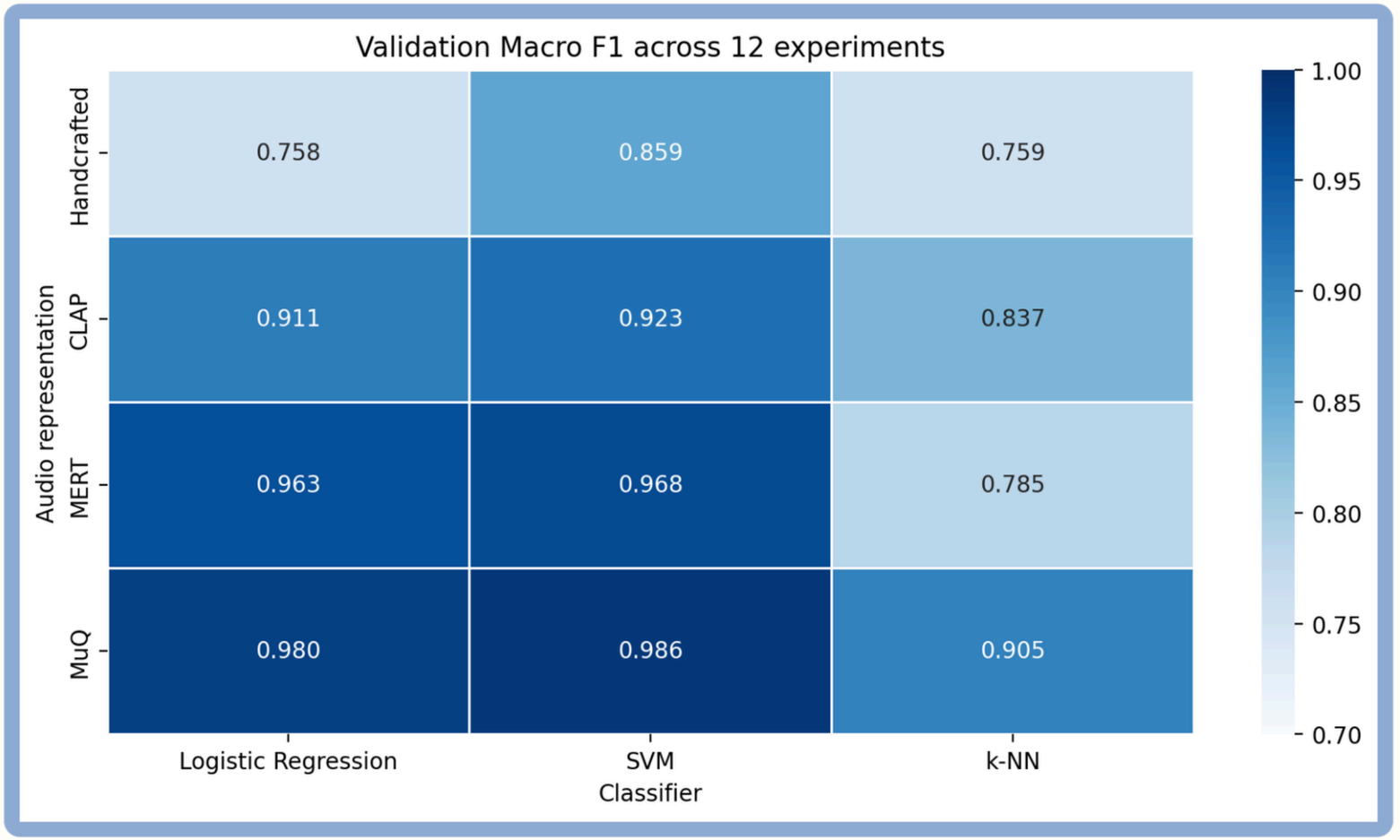
CLAP | MERT | MuQ EMBEDDINGS



- UMAP reveals some generator-specific structure, but the classes are not perfectly separated in two dimensions.

BASELINE RESULTS

06



Evaluation metrics

- Accuracy
- Balanced Accuracy
- Macro Precision
- Macro Recall
- Macro F1
- Confusion Matrix

Music-focused embeddings substantially improved closed-set generator classification.
MuQ + SVM achieved the best test **Macro F1 of 0.9896.**

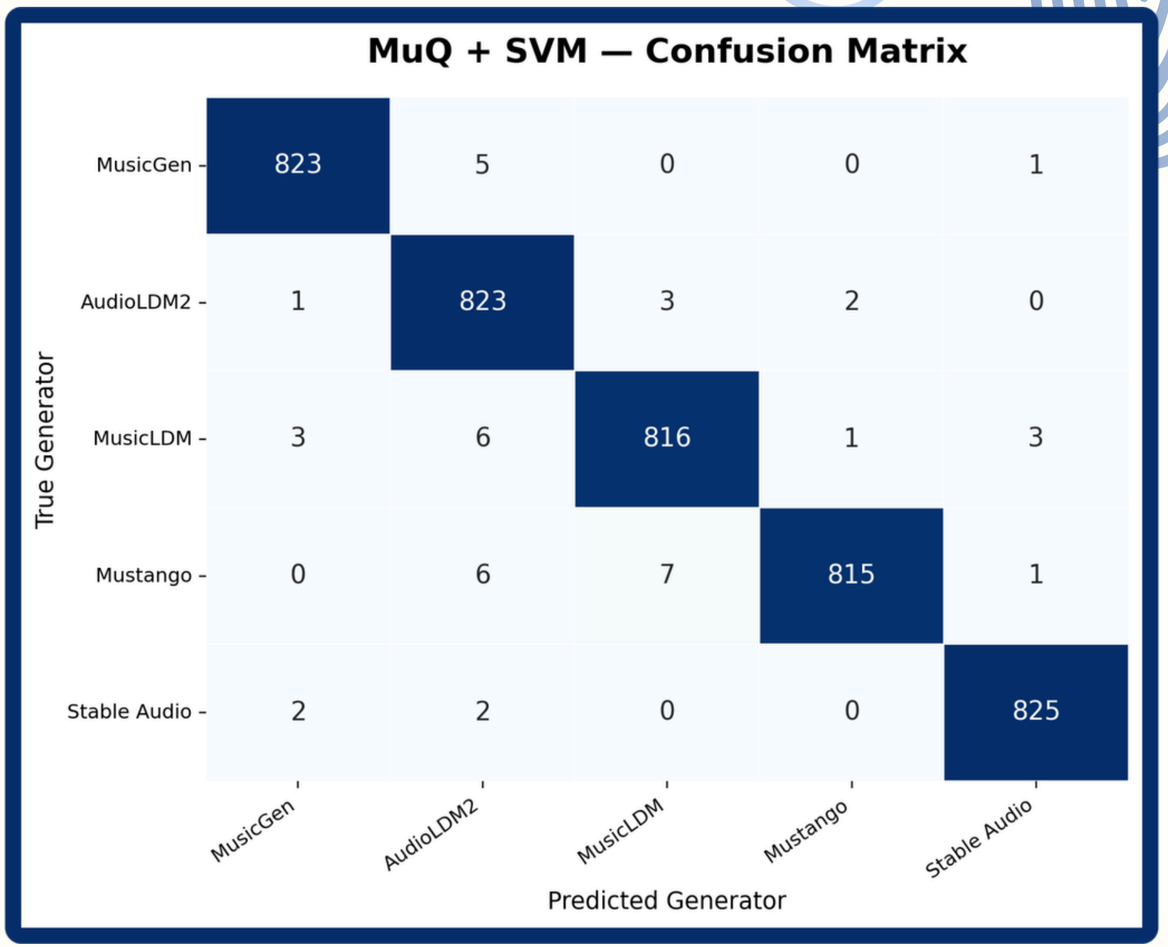
07

ERROR ANALYSIS

Number of test errors per top-5 models (observed on 4145 test clips):

- CLAP + SVM: 344 (8,30%)
- MERT + SVM: 197 (4,75%)
- MERT + LR: 181 (4,37%)
- MuQ + LR: 68 (1,64%)
- **MuQ + SVM: 43 (1,04%)**

The most frequent mistake is classifying **Mustango** as **MusicLDM**.



PROGRESS VS PROPOSAL

COMPLETED

- FakeMusicCaps audit
- Handcrafted features + frozen embeddings
- Representation analysis with PCA / UMAP
- LR, SVM and k-NN baseline classification
- Closed-set evaluation and error analysis

TO DO

- AI vs real classification
- Unseen-generator testing
- Cross-dataset evaluation
- Quantitative clustering analysis

Current Limitation

Our results demonstrate strong closed-set classification on FakeMusicCaps, but generalization to real music, unseen generators, and other datasets remains untested.

NEXT STEPS & PLAN B

01

AI vs Real

02

Unseen Generator

03

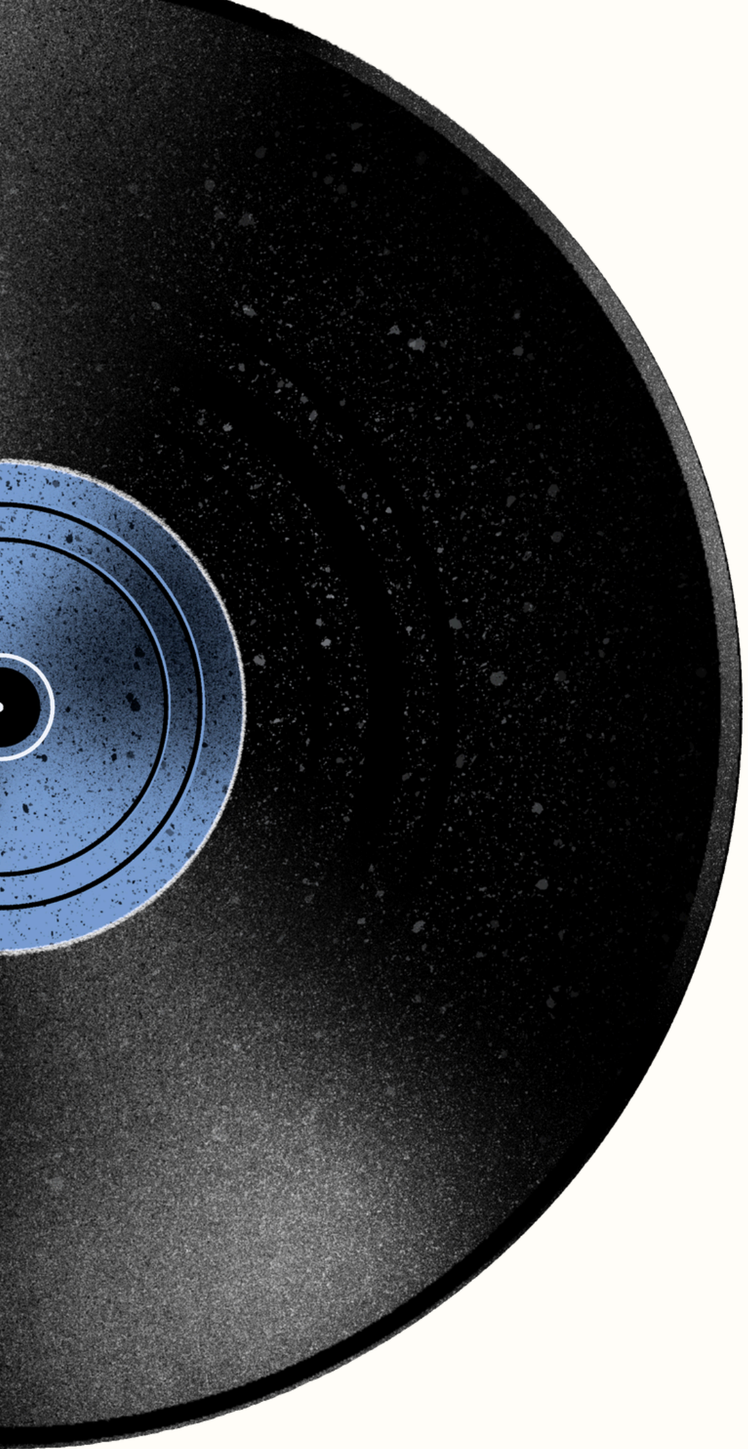
Cross-Dataset Generalization

RISK / PLAN B

If performance drops on unseen generators, investigate domain shift, analyze failure cases, and explore unknown-class detection.

ADDITIONAL EXPERIMENTS

Explore feature fusion (MuQ + handcrafted / MERT) to improve generalization. Analyze errors by musical content, including vocal vs non-vocal tracks.



THANK YOU!

For Your Attention

